

What is CSS?

CSS (Cascading Style Sheets) is a stylesheet language used to style and design HTML elements on a web page.

HTML defines the structure and content of a webpage, while CSS controls its appearance and layout.

Ways to Add CSS

CSS can be added to HTML in three ways:

1. Inline CSS

CSS is written directly inside the HTML element using the style attribute for applying a style to a single element.

style="css1: value; css2 : value;"

```
<h1 style="color: blue;">Hello World</h1>
```

2. Internal CSS

CSS is written inside the <style> </style> tag, usually within the <head></head> section for styling a single HTML page.

```
<head>
  <style>
    h1 {
      color: blue;
    }
  </style>
</head>
```

3. External CSS

CSS is written in a separate **.css** file and connected to HTML using the <link> tag.

Best approach for most real-world websites because the same CSS file can be used across multiple HTML pages.

— **index.html (HTML FILE)** —

```
<head>
  <link rel="stylesheet" href="style.css">
</head>
```

— **style.css (CSS FILE)** —

```
h1 {
  color: blue;
}
```

CSS Selectors

A CSS Selector is used to select the HTML element, that we want to style.

Syntax:

```
selector {  
    property: value;  
}
```

Types of CSS Selectors

1. Element Selector :

Element Selector selects all HTML elements of a specific type or tag.

```
ex.:  p {  
        color: blue;  
    }
```

2. ID Selector

ID Selector selects an HTML element using its unique id attribute using (#) symbol.

```
ex.:  #title {  
        color: red;  
    }
```

```
<h1 id="title">Apply red color only in this heading tag</h1>
```

3. Class Selector

Class Selector selects one or more HTML elements having the same class name using (.) symbol.

```
ex.:  .text {  
        color: green;  
    }
```

```
<h1 class="title">Apply red color only in this heading tag</h1>
```

```
<h3 class="title">Apply red color only in this heading tag</h3>
```

4. Universal Selector

Universal Selector selects all HTML elements on the page using (*) symbol.

```
ex.:  * {  
        margin: 0;  
        padding: 0;  
    }
```

5. Grouping Selector

```
h1, h2, p {color: blue; }
```

SOME CSS PROPERTIES

1. CSS Colors

color : red **or** rgb(255, 255, 0) **or** #FEDA00;

background-color

border-color

Color formats: (Name, RGB, RGBA, HEX)

4. CSS Backgrounds

Background properties control the colour and image displayed behind an element.

Properties / Values:

- background-color → red or #ff0000 or rgb(255, 0, 0)
- background-image → url("image.jpg") or linear-gradient(...)
- background-repeat → repeat or no-repeat or repeat-x or repeat-y
- background-position → left or center or right or top or bottom or 50% 50%
- background-size → auto or cover or contain or 100% 100%
- background-attachment → scroll or fixed or local
- background → Shorthand for multiple background properties

5. CSS Borders

Borders create a visible boundary around an element.

Properties / Values:

- border-style → solid or dashed or dotted or double or none
- border-width → 1px or 2px or thin or medium or thick
- border-color → red or #ff0000 or rgb(...)
- border-top → 2px solid red
- border-right → 2px solid red
- border-bottom → 2px solid red
- border-left → 2px solid red
- border-radius → 10px or 50%
- border → 1px solid black

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7. CSS Text

Text properties control alignment, decoration, transformation and spacing.

Properties / Values:

- `text-align` → `left` or `center` or `right` or `justify`
- `text-decoration` → `none` or `underline` or `overline` or `line-through`
- `text-transform` → `none` or `uppercase` or `lowercase` or `capitalize`
- `text-indent` → `20px` or `2em`
- `line-height` → `normal` or `1.5` or `24px`
- `letter-spacing` → `normal` or `2px` or `0.1em`
- `word-spacing` → `normal` or `5px` or `0.2em`
- `text-shadow` → `2px 2px 5px gray`

9. CSS Margin & Padding

Margin creates space outside an element, while **padding** creates space inside it.

Properties / Values:

- `margin` → `20px` or `10px 20px` or `10px auto`
- `margin-top` → `20px` or `2rem`
- `margin-right` → `20px` or `2rem`
- `margin-bottom` → `20px` or `2rem`
- `margin-left` → `20px` or `2rem`
- `padding` → `20px` or `10px 20px`
- `padding-top` → `20px` or `2rem`
- `padding-right` → `20px` or `2rem`
- `padding-bottom` → `20px` or `2rem`
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